



Map courtesy of Wikipedia, created by San Jose. Used with permission.

Backtracking
Search
and
Constraint
Satisfaction



<-- That is
Europe

Pretty...

Notice the
map uses
6 colours

but

we can do
better!

Map courtesy of Wikipedia, created by San Jose. Used with permission.

- We can colour the map using only four colours in such a way that no adjacent countries have the same colour
- To do so, we formulate the map colouring problem as graph, with nodes representing countries, and edges that represent adjacency

Q1 – Draw one such graph for the following countries:

Portugal, Spain, Andorra, France, Luxembourg, Belgium, Netherlands, Germany, Switzerland, Italy, Czech Republic, Austria, and San Marino

Be careful that your graph does not contain crossing edges!

- We can find a suitable colouring using backtracking search. Assuming that you can use the four colours **red, green, blue, yellow**. Answer the following:

Q2 – What are the variables in this problem? What do they represent?

Q3 – What is the domain of each variable?

Q4 – Which variable(s) intervene in the largest number of constraints? (and how many?)

Q5 – Which variable(s) intervene in the least number of constraints? (and how many?)

-Recall there are rules to help you decide which variables to choose earlier during the search.

Q6 – In this graph, which variable should be chosen first?

Q7 – Which variable should be chosen next?

Q8 – Show the first 3 levels of the search tree with the values assigned to the variables chosen at each level (note we're not actually searching, since the search would go all the way down before backtracking).

Weekly meta-question...

Is the current combination of lecture time, programming assignments, and Programming exercises helping you gain a solid understanding of the AI topics covered so far?

(yes, no, and why. Also, if your answer is no, what is missing from the course in your opinion?)

Please complete this problem set, and submit on Quercus